

Discrete Math — Monday Test

Theoretical Questions — Worked Solutions

Question numbers (T1–T18) match the companion Theoretical Questions PDF exactly.

Tuesday

T1. a) The negation of a proposition p , written $\neg p$, is the proposition “it is not the case that p ” — true exactly when p is false, and false exactly when p is true.

b) “This is not a boring course.”

T2. a)

p	q	$p \vee q$	$p \wedge q$	$p \oplus q$	$p \rightarrow q$	$p \leftrightarrow q$
T	T	T	T	F	T	T
T	F	T	F	T	F	F
F	T	T	F	T	T	F
F	F	F	F	F	T	T

b) Let p : “I’ll go to the movies tonight,” q : “I’ll finish my discrete mathematics homework.”

- $p \vee q$: “I’ll go to the movies tonight or I’ll finish my discrete mathematics homework (or both).”
- $p \wedge q$: “I’ll go to the movies tonight and I’ll finish my discrete mathematics homework.”
- $p \oplus q$: “I’ll go to the movies tonight or I’ll finish my discrete mathematics homework, but not both.”
- $p \rightarrow q$: “If I go to the movies tonight, then I’ll finish my discrete mathematics homework.”
- $p \leftrightarrow q$: “I’ll go to the movies tonight if and only if I finish my discrete mathematics homework.”

T3. a) Common English phrasings of $p \rightarrow q$: “if p then q ”; “ p implies q ”; “ p only if q ”; “ q if p ”; “ q whenever p ”; “ p is sufficient for q ”; “ q is necessary for p .” (Any five of these.)

b) **Converse:** $q \rightarrow p$. **Contrapositive:** $\neg q \rightarrow \neg p$.

c) Statement: “If it is sunny tomorrow, then I will go for a walk in the woods.”

Converse: “If I go for a walk in the woods, then it is sunny tomorrow.”

Contrapositive: “If I do not go for a walk in the woods, then it is not sunny tomorrow.”

T4. a) Two propositions are **logically equivalent** if they have the same truth value in every possible row of a truth table — equivalently, their biconditional is a tautology.

b) Ways to show equivalence: (i) build a truth table for both sides and check every row matches;

(ii) start from one side and apply a chain of known logical identities (De Morgan's, distributive, conditional identity, etc.) until you reach the other side.

c) Show $\neg p \vee (r \rightarrow \neg q) \equiv \neg p \vee \neg q \vee \neg r$:

Method 1 (identity chain): $r \rightarrow \neg q \equiv \neg r \vee \neg q$ (using $a \rightarrow b \equiv \neg a \vee b$). So $\neg p \vee (r \rightarrow \neg q) \equiv \neg p \vee (\neg r \vee \neg q) \equiv \neg p \vee \neg q \vee \neg r$ by associativity/commutativity. ■

Method 2 (truth table logic): The right side is false only when $p = q = r = \text{T}$. The left side is false only when $\neg p = \text{F}$ (so $p = \text{T}$) and $r \rightarrow \neg q = \text{F}$ (so $r = \text{T}, q = \text{T}$) — the same single case. Since both sides are false in exactly the same case, they're equivalent.

T5. a) For each row of the truth table where the compound proposition is True, build a **minterm**: an AND of all the variables, using the variable itself if it's True in that row and its negation if it's False. Then OR all these minterms together — this reproduces exactly the given truth table. This is **disjunctive normal form (DNF)**.

b) Since *any* truth table can be reproduced this way using only $\wedge$, $\vee$, and $\neg$, these three operators together can express every possible Boolean function — that's exactly the definition of **functionally complete**.

c) Yes: the **NAND** operator ($\uparrow$) alone is functionally complete, and so is **NOR** ($\downarrow$) alone.

Wednesday

T6. $\forall x P(x)$ is true when $P(x)$ holds for *every* x in the domain. $\exists x P(x)$ is true when $P(x)$ holds for *at least one* x . Their negations (De Morgan's for quantifiers):

$$\neg \forall x P(x) \equiv \exists x \neg P(x), \quad \neg \exists x P(x) \equiv \forall x \neg P(x).$$

Thursday

T7. a) $\exists x \forall y P(x, y)$ says there is *one single* x that works for every y — a “champion” x . $\forall y \exists x P(x, y)$ only says that for each y *some* x works, but that x is allowed to change depending on y . The first statement is strictly stronger: it implies the second, but not conversely.

b) Let the domain be all integers and $P(x, y)$: “ $x + y = 0$.” Then $\exists x \forall y P(x, y)$ is **false** (no single x satisfies $x + y = 0$ for every y), but $\forall y \exists x P(x, y)$ is **true** (given any y , choose $x = -y$). Different truth values, as required.

T8. A **valid argument** is one where the conclusion must be true whenever all the premises are true — i.e., the argument form is a tautology. Let p : “the earth is flat,” q : “you can sail off the edge of the earth.” The premises are $p \rightarrow q$ and $\neg q$; the conclusion is $\neg p$. This is exactly **Modus Tollens** ($p \rightarrow q, \neg q \therefore \neg p$), a known valid inference rule — so the argument is valid.

Friday

T9. Let $Z(x)$: “ x is a zebra,” $S(x)$: “ x has stripes.” Premises: $\forall x (Z(x) \rightarrow S(x))$ and $Z(\text{Mark})$.

Universal Instantiation at $x = \text{Mark}$: $Z(\text{Mark}) \rightarrow S(\text{Mark})$.

Modus Ponens with $Z(\text{Mark})$: $S(\text{Mark})$ — “Mark has stripes.” ■

Saturday

T10. a) **Direct proof:** assume p is true, and derive q through a chain of valid steps. **Proof by contraposition:** assume $\neg q$, and derive $\neg p$ (valid since $p \rightarrow q \equiv \neg q \rightarrow \neg p$). **Proof by contradiction:** assume p true and q false simultaneously, and derive a logical contradiction — this forces $p \rightarrow q$ to be true.

b) Statement: “If n is even, then $n + 4$ is even.”

- *Direct:* n even means $n = 2k$. Then $n + 4 = 2k + 4 = 2(k + 2)$, which is even.
- *Contraposition:* Assume $n + 4$ is odd, say $n + 4 = 2k + 1$. Then $n = 2k - 3 = 2(k - 2) + 1$, which is odd — this is the contrapositive “ $n + 4$ odd $\Rightarrow n$ odd,” equivalent to the original statement.
- *Contradiction:* Assume n is even and $n + 4$ is odd. Since $n = 2k$, $n + 4 = 2(k + 2)$ is even — contradicting the assumption that $n + 4$ is odd. So whenever n is even, $n + 4$ must be even.

T11. a) To prove $p \leftrightarrow q$, prove both directions separately: $p \rightarrow q$ and $q \rightarrow p$ (or show they’re always simultaneously true or simultaneously false via a chain of equivalences).

b) Statement: “ $3n + 2$ is odd if and only if $9n + 5$ is even.”

($\rightarrow$) Suppose $3n + 2$ is odd. Then $3n$ is odd (odd – even = odd), which forces n to be odd (if n were even, $3n$ would be even). Write $n = 2k + 1$: then $9n + 5 = 9(2k + 1) + 5 = 18k + 14 = 2(9k + 7)$, which is even.

($\leftarrow$) Suppose $9n + 5$ is even. Then $9n$ is odd (even – odd = odd), which forces n to be odd. Write $n = 2k + 1$: then $3n + 2 = 3(2k + 1) + 2 = 6k + 5 = 2(3k + 2) + 1$, which is odd.

Both directions hold, so the biconditional is proved. ■

T12. No, it is not sufficient. The implications $p_4 \rightarrow p_2$, $p_3 \rightarrow p_1$, and $p_1 \rightarrow p_2$ don’t connect all four statements into a single cycle — there’s no path leading back out of p_2 , so you can never conclude anything flows from p_2 to the others. To prove all four equivalent, you need one single cycle of implications touching every statement, for example:

$$p_1 \rightarrow p_2, \quad p_2 \rightarrow p_3, \quad p_3 \rightarrow p_4, \quad p_4 \rightarrow p_1.$$

T13. a) To disprove $\forall x P(x)$, it’s enough to find just **one counterexample** — a specific value c in the domain for which $P(c)$ is false.

b) Statement: “For every positive integer n , $n^2 \geq 2^n$.” **Counterexample:** $n = 5$. $5^2 = 25$ but $2^5 = 32$, and $25 \geq 32$ is false. So the statement is false.

T14. A **constructive** existence proof explicitly exhibits an object with the desired property. A **nonconstructive** proof shows such an object must exist (often via contradiction, counting, or the pigeonhole principle) without ever identifying it.

Constructive example: “There is an even prime number” — proof: 2 is prime and even, done.

Nonconstructive example: “Either $2 \cdot 10^{500} + 15$ or $2 \cdot 10^{500} + 16$ is not a perfect square” — since consecutive integers can never both be perfect squares, one of them must fail, but the proof never says which.

T15. A uniqueness proof for “there is a unique x such that $P(x)$ ” has two parts:

1. **Existence:** show some element x with $P(x)$ true exists.
2. **Uniqueness:** show that if $P(y)$ is also true for some y , then y must equal x — i.e., any two elements satisfying P are actually the same element.

T16. Split into an **exhaustive set of cases** based on the signs of x and y (four cases: both nonnegative, x nonnegative & y negative, x negative & y nonnegative, both negative). In each case, use the sign-specific definition of $|\cdot|$ to compute both $|xy|$ and $|x||y|$ directly and confirm they match. For example, if $x \geq 0, y < 0$: $xy \leq 0$ so $|xy| = -xy = x(-y) = |x||y|$. Since the four cases cover every possible pair of reals, the equality holds in general.

Sunday

T17. a) **Yes** — once you already have a candidate formula (often found by spotting a pattern in small cases), induction can prove it holds for all n .

b) **Yes** — induction can confirm whether a proposed formula is correct: verify the base case and the inductive step; if both hold, the formula is correct for all n . (It cannot help you *discover* the formula — only confirm one you already have.)

c) Formula: $2 + 4 + \cdots + 2n = n(n + 1)$.

Base case $n = 1$: LHS = 2, RHS = $1 \cdot 2 = 2$. ✓

Inductive hypothesis: assume $2 + 4 + \cdots + 2k = k(k + 1)$ for some positive integer k .

Inductive step: $2 + 4 + \cdots + 2k + 2(k + 1) = k(k + 1) + 2(k + 1) = (k + 1)(k + 2)$, matching the formula at $n = k + 1$. By induction, true for all positive integers n . ■

T18. a) Check small values: $n = 1$ through 6 all fail ($11n + 17$ grows faster than 2^n at first). At $n = 7$: $11(7) + 17 = 94 \leq 2^7 = 128$ — **true**. Checking $n = 8$: $105 \leq 256$ — also true. **Conjecture:** $11n + 17 \leq 2^n$ **for all integers** $n \geq 7$.

b) **Base case** $n = 7$: $94 \leq 128$. ✓

Inductive hypothesis: assume $11k + 17 \leq 2^k$ for some integer $k \geq 7$.

Inductive step:

$$11(k + 1) + 17 = (11k + 17) + 11 \leq 2^k + 11.$$

Since $k \geq 7$, $2^k \geq 128 \geq 11$, so $2^k + 11 \leq 2^k + 2^k = 2^{k+1}$. Chaining:

$$11(k + 1) + 17 \leq 2^k + 11 \leq 2^{k+1}.$$

By induction, $11n + 17 \leq 2^n$ holds for all integers $n \geq 7$. ■